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Sampling microcanonical ensembles of trajectories using harmonic approximation in internal coordinates

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In this paper, we modify quasiclassical harmonic sampling of microcanonical ensembles of trajectories by using the curvilinear internal coordinates. The harmonic approximation in the curvilinear normal coordinates provides a more realistic description of the PES than in the conventional rectilinear ones at finite displacements. Therefore, the sampling of vibrations in the internal coordinates significantly improves the quality of the sampling in a block-box fashion, providing more realistic displacements and reducing the errors in the potential energy. In particular, the sampling of large-amplitude torsion vibrations, which is non-realistic in the Cartesian modes, becomes accurate or acceptable in the curvilinear modes. © 2014 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4892109>]

I. INTRODUCTION

Integrating microcanonical (NVE) ensembles of classical trajectories is an important and widely used tool to study the dynamics of gas-phase reactions. Most applications are performed by means of direct Born-Oppenheimer molecular dynamics with forces calculated on-the-fly by electronic structure theory methods.¹ A “swarm” of trajectories is usually integrated for the proper statistical averaging. Since NVE conditions require that the system is not interacting with the surroundings, no equilibration stage is present in the simulations. The results thus strongly depend on the initial conditions: coordinates and velocities. To generate realistic trajectories, one needs to have a good approximation to the initial state wave function or its classical analogue.² For adiabatic dynamics, a complete set of rotational-vibrational wave functions for the selected electronic state is required. However, accurate solution for the time-independent nuclear problem is computationally as demanding as trajectory simulations. Therefore, a simple “harmonic oscillator-rigid rotor” approximation is routinely used to obtain the starting conditions. In this paper, we restrict ourselves to sampling molecular vibrations.

The sampling procedure involves the following steps:^{3–5}

- Find a stationary point $\mathbf{x}_0$ on the potential energy PES, calculate the molecular Hessian, $\mathbf{F}$.
- Solve the vibrational problem in the harmonic approximation.
- Sample quantum numbers (vibrational levels and, consequently, energies) according to the assumed energy distribution.
- Sample displacements and momenta (velocities) in normal coordinates.

- Transform displacements and velocities to the Cartesian frame, $\Delta\mathbf{x}$ and $\mathbf{v}$, calculate the harmonic potential, V^{harm} , and kinetic energies in the harmonic approximation.
- Calculate the Born-Oppenheimer potential energy $\mathbf{x}_0 + \Delta\mathbf{x}$, V , by electronic structure theory.
- Scale $\Delta\mathbf{x}$ and so that $V^{\text{harm}} = V$.

The first two steps are performed once, whereas the others are repeated for each trajectory in the set. The scaling in the final step is meant to introduce the corrections for anharmonicity which inevitably comes out for finite displacements from the stationary point. Although this procedure gives the correct energy (macrostate), it distorts the microstate since coordinates are scaled uniformly although their contributions to anharmonicity are different.

It is thus desirable to reduce sampling errors without going beyond the harmonic approximation. In this paper, we are going to show that it is possible by solving the harmonic vibrational problem in curvilinear internal coordinates—bond lengths, valence, and torsion angles, etc. Being chemically natural they are known to result in smaller couplings and anharmonicities than the Cartesian coordinates do.

II. THEORY

A. Curvilinear internal coordinates

First, we define the system of internal coordinates $\mathbf{q}(\mathbf{x})$ which are explicit functions of the Cartesian coordinates $\mathbf{x}$. In general case, $\mathbf{q}$ may be a redundant set. The Jacobian used for transformation between the two systems is known as the $\mathbf{B}$ -matrix: $B_{ij} = \frac{\partial q_i}{\partial x_j}$.⁶ If $\mathbf{q}$ are redundant then the problem should be solved using a non-redundant set $\mathbf{s}$, which spans the range of $\mathbf{B}$ and is defined by some linear transformation:

$$\mathbf{s} = \mathbf{U}'\mathbf{q}. \quad (1)$$

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In this work, we use the redundant set of primitive internal coordinates^{7,8}—bond lengths, valence, and torsion angles— $\mathbf{q}$ —and the non-redundant coordinates— $\mathbf{s}$ —are delocalized internal coordinates.⁹ They are found by singular value decomposition of $\mathbf{B}$ and $\mathbf{U}$ is composed of non-redundant right-singular vectors. The Jacobian between $\mathbf{x}$ and $\mathbf{s}$ is now

$$\mathbf{B}_s = \mathbf{U}^t \mathbf{B}. \quad (2)$$

B. Harmonic vibrations in internal coordinates

To sample the trajectories, we need to solve the vibrational harmonic problem in internal coordinates. These have been accomplished in several publications, namely, in Refs. 10–13 and a more accurate approach by Sipachev.¹⁴ Here we use the formulation of Jensen and Palmer,¹³ according to which the harmonic vibrational problem in curvilinear internal coordinates is solved by the following secular equation:

$$(\mathbf{G}_m^{1/2} \mathbf{F}_s \mathbf{G}_m^{1/2}) \mathbf{V} = \mathbf{V} \mathbf{\Lambda}. \quad (3)$$

In this equation, $\mathbf{F}_s$ is the force constant matrix (molecular Hessian) in delocalized internal coordinates, which is for the special case of PES extremum defined as follows:

$$\mathbf{F}_s = (\mathbf{B}_s^t)^+ \mathbf{F} \mathbf{B}_s^+, \quad (4)$$

and $\mathbf{G}_m$ is the kinetic energy matrix:

$$\mathbf{G}_m = \mathbf{B}_s \mathbf{M}^{-1} \mathbf{B}_s^t, \quad (5)$$

where $\mathbf{M}$ is a diagonal matrix of nuclear masses. Here, $+$ denotes the Moore-Penrose pseudoinverse. Note that, unlike for the harmonic problem in Cartesian coordinates, the kinetic operator $\mathbf{G}_m$ is not exact since nonlinear terms arising upon coordinate transformation are truncated.

Vibrational frequencies are defined by the eigenvalues $\mathbf{\Lambda}$ and curvilinear normal coordinates $\mathbf{L}$ are found as

$$\mathbf{L} = \mathbf{G}_m^{1/2} \mathbf{V}. \quad (6)$$

C. Sampling

As soon as vibrational frequencies and normal coordinates are found, sampling is performed. Quantum numbers are selected manually or found by a procedure such as the von Neumann rejection technique.¹⁵ Once that has been done, displacements and velocities may be sampled either in a classical^{3,4} or in a quantum⁴ manner using Wigner¹⁶ or else Husimi^{17,18} functions.

In this paper, we sample the Boltzmann distribution for each of the modes at a certain temperature and find canonical coordinates and momenta for the i th mode classically as follows:

$$Q_i = (2E_i/f_i)^{1/2} \cos 2\pi R_i, \quad (7)$$

$$P_i = (2E_i)^{1/2} \sin 2\pi R_i, \quad (8)$$

where E_i is the energy of the harmonic vibration in the selected state, f_i is the corresponding vibrational eigenvalue, and

R_i is a newly generated random number uniformly distributed on $[0, 1]$. The classical vibrational amplitude is

$$A_i = (2E_i/f_i)^{1/2}. \quad (9)$$

D. Internal to Cartesian-back transformation

Once displacements are defined, they need to be transformed to the Cartesian frame, since Cartesian coordinates are required to calculate electronic energy. Displacements in non-redundant internal coordinates are obtained trivially:

$$\Delta \mathbf{s}(\mathbf{Q}) = \mathbf{L} \mathbf{Q}. \quad (10)$$

To find Cartesian displacements $\Delta \mathbf{x}(\mathbf{Q})$ corresponding to $\Delta \mathbf{s}(\mathbf{Q})$ is more difficult, since the latter are nonlinear functions of the former. Nevertheless, internal to Cartesian back-transformation can be performed up to a given accuracy by several methods: iterative back-transformation,¹⁹ simplified iterative back-transformation,^{20,21} transformation using auxiliary Z-matrix,²⁰ and the method using reciprocal vector bases.¹⁰ Here we apply the recently developed high-order path expansion (HOPE) back-transformation,²² using high-order geometric derivatives of the internal coordinates with respect to the Cartesian ones, computed by automatic differentiation techniques.

E. Scaling

If $V^{\text{harm}}(\Delta \mathbf{x})$ and $V(\Delta \mathbf{x})$ are the potential energy in the harmonic approximation, and the true Born-Oppenheimer potential energy at finite displacement $\Delta \mathbf{x}$, respectively, then the following scaling:

$$\mathbf{x}' = k^{1/2} \mathbf{x}, \quad (11)$$

$$k = \frac{V^{\text{harm}}(\Delta \mathbf{x})}{V(\Delta \mathbf{x})}, \quad (12)$$

leads to $V^{\text{harm}}(\Delta \mathbf{x}') = V(\Delta \mathbf{x})$. Velocities are scaled accordingly to preserve the equality of the potential and kinetic energies in the harmonic approximation:

$$\mathbf{v}' = k^{-1/2} \mathbf{v}. \quad (13)$$

Such scaling assumes that k is close to 1 and is meant to make starting conditions more realistic.

Here we introduce two scaling coefficients depending on how the finite displacement has been generated: using rectilinear normal modes, k_x , or curvilinear ones, k_s .

III. COMPUTATIONAL CONSIDERATIONS

From computational point of view, the most expensive procedure in this scheme is the internal-to-Cartesian back transformation,²² which is to be performed once for each trajectory. The cost of the transformation itself is dominated by the cost of inverting the Wilson's $\mathbf{B}$ matrix. Thus, the use of the HOPE²² transformation, requiring only one inversion for $\mathbf{B}(\mathbf{x}_0)$ for all M trajectories in the ensemble, will not add any significant computational time to the procedure. By contrast, conventional iterative back-transformation¹⁹ requires on

average 5–10 matrix inverses to transform $\Delta \mathbf{s}_i$ into $\Delta \mathbf{x}_i$, that is, from $5M$ to $10M$ inversions are to be computed for each member of the whole ensemble. Consequently, sampling may take non-negligible time compared to electronic structure calculations, especially for big M . Indeed, integrating hundreds of trajectories is not uncommon in applications (see Ref. 1).

IV. RESULTS AND DISCUSSION

We test our algorithm on the acetone molecule $\text{CH}_3\text{--CO--CH}_3$ in its equilibrium geometry where the force constant matrix has been calculated and the harmonic vibrational problem has been solved. First, to probe the difference between curvilinear and rectilinear normal coordinates we perform energy scans along the normal modes. After that, we go on to sample the ensembles of trajectories at various temperatures.

All electronic structure calculations have been performed using DALTON^{24,26} and LSDALTON^{25,26} programs at DFT/BLYP^{27,28}/6-311G**²⁹ level of theory and the sampling has been made with a Python script. To perform HOPE back-transformation automatic differentiation library libtaylor²³ has been applied.

A. PES scans

Solving the harmonic problem for acetone in Cartesian and internal coordinates results in vibrational frequencies identical within fractions of cm^{-1} . In Figs. 1 and 2 we have plotted energy scans along the three lowest-frequency modes at the given level of electronic structure theory: $\nu_1 = 23 \text{ cm}^{-1}$ corresponding to conrotatory torsion of methyl groups, $\nu_2 = 133 \text{ cm}^{-1}$ corresponding to disrotatory torsion of methyl groups, and $\nu_3 = 368 \text{ cm}^{-1}$ which is the deformation of carbon skeleton of the molecule. The range of displacement is

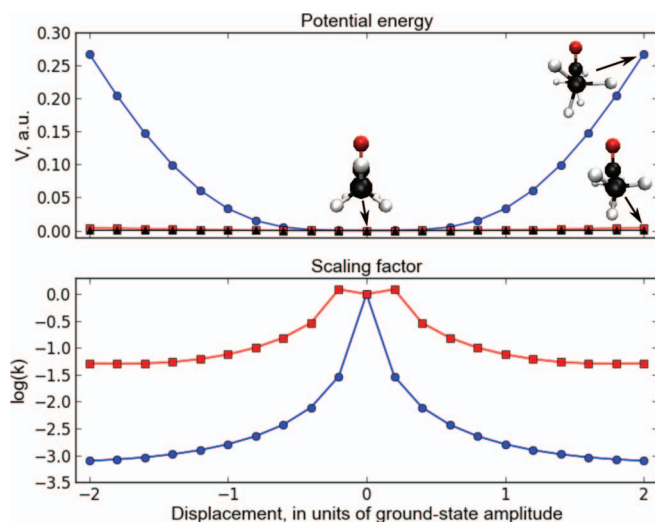


FIG. 1. Torsional vibration ν_1 : PES scan with illustrations of geometric displacements (top) and scaling coefficients (bottom). Black—harmonic potential; blue—scan in rectilinear normal coordinates and k_x ; and red—scan in curvilinear normal coordinates and k_s .

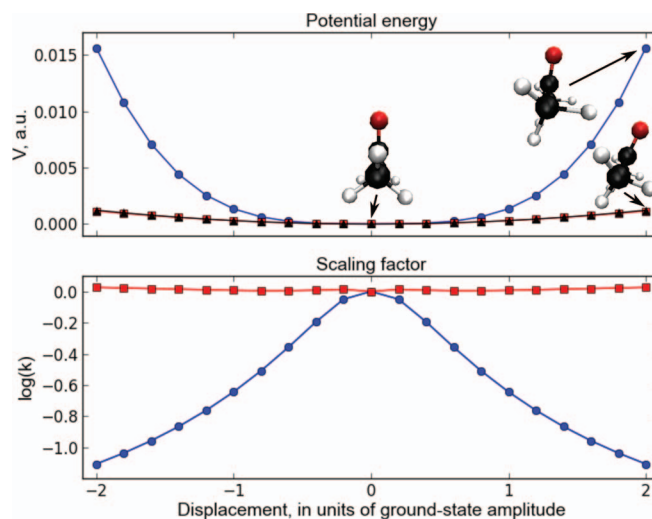


FIG. 2. Torsional vibration ν_2 : PES scan with illustrations of geometric displacements (top) and scaling coefficients (bottom). Black—harmonic potential; blue—scan in rectilinear normal coordinates and k_x ; and red—scan in curvilinear normal coordinates and k_s .

extended up to two classical ground-state amplitudes A_i , as the regions of space beyond A_i are accessible for excited states in case of classical sampling and for any state in case of quantum-mechanical sampling. From Figs. 1 and 2 it becomes immediately apparent that harmonic approximation in curvilinear internal coordinates is a much better one than in Cartesian. In case of both, torsional vibrations for finite displacements from the equilibrium, the latter fails almost immediately.

For ν_1 , although k_s drops to the value of 0.1 and more, it is everywhere about two orders of magnitude smaller than k_x . For ν_2 in the range under consideration there is essentially no anharmonicity in curvilinear coordinates and k_s is everywhere close to one. Unlike this, rectilinear harmonic approximation

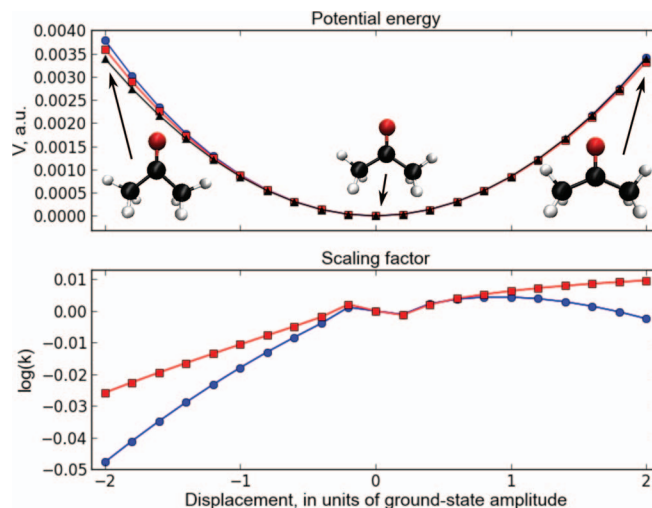


FIG. 3. Deformational vibration ν_3 : PES scan with illustration of geometric displacements (top) and scaling coefficients (bottom). Black—harmonic potential; blue—scan in rectilinear normal coordinates and k_x ; and red—scan in curvilinear normal coordinates and k_s . The difference between Cartesian and internal coordinate step geometries is visually indistinguishable on the scale of the figure, and only the Cartesian step geometry is shown.

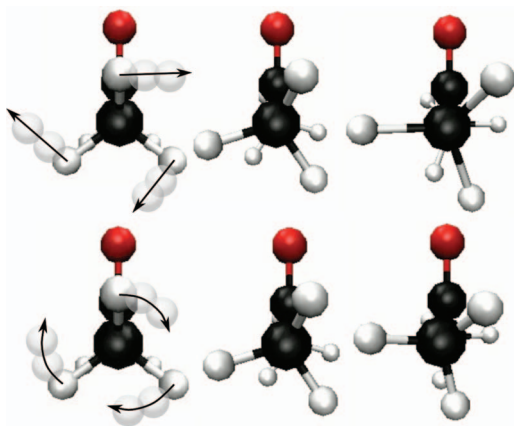


FIG. 4. Illustration of the difference between normal mode following in Cartesian coordinates (top), and internal coordinates (bottom) for the lowest torsional mode of acetone (motion shown in one direction only).

again fails to represent the true section of the PES. Therefore, by using curvilinear internal coordinates, the harmonic approximation is significantly improved in the case of ν_1 and becomes qualitatively accurate in case of ν_2 as compared to conventional rectilinear normal coordinates.

For the deformation ν_3 (see Fig. 3) both harmonic curves follow the true PES quite closely, with the curvilinear normal mode scan being slightly more accurate for folding the C–C–C skeleton and the rectilinear normal mode scan being better for unfolding. Nevertheless, on average the curvilinear description is slightly preferable. Other vibrations with higher frequencies do not exhibit any noticeable differences between the representations.

The reason for the complete failure of the harmonic approximation in the Cartesian coordinates in case of ν_1 and ν_2 is the rapid increase of the coupling between them and the C–H bonds stretching at finite displacements. If internal coordinates are used, the contribution of this coupling is also significant for the lower-frequency torsion, but is already negligible for the higher-frequency vibration. These statements are illustrated in Fig. 4. It is clearly seen that for ν_1 C–H bonds are significantly stretched in case of rectilinear-mode scan. In fact, at $2A_1$ they are elongated ca. 50% with respect to the equilibrium length. The analogous elongation for ν_2 is about 25%. In case of the curvilinear normal modes, C–H elongation is much smaller for ν_1 and insignificant for ν_2 . Such dramatic effects of coordinate system choice become apparent for large-amplitude vibrations involving angular coordinates, that is, low-frequency ones such as methyl group torsions.

B. Sampling trajectories

After having ensured that the use of curvilinear internal coordinates indeed improves the harmonic description of large-amplitude vibrations in acetone, we have sampled three sets of 300 trajectories each from the Boltzmann distribution at three temperatures: 300 K, 500 K, and 700 K. To facilitate the comparison, we used the same vibrational quantum numbers and random phases in Eq. (7) for both curvilinear and rectilinear-mode sampling. The results are given in Table I.

It is readily seen from these data that using curvilinear normal coordinates reduces both accumulated and maximal error in potential energy substantially. RMS deviation of scaling coefficient is also reduced by about a factor of two. Average values of k are less representative as measures of error, since k can be both greater and smaller than 1.0, leading to error cancellation after averaging. Nevertheless, average values of the scaling factors are instructive: the fact that they are in all cases smaller than 1.0, especially for the rectilinear modes, points at the dominant source of errors—sampling of torsional vibrations. Indeed, as we have seen from Figures 1 and 2, the harmonic approximation underestimates the potential energy, and the maximal errors for the sampling in the rectilinear normal coordinates correlate quite well with those for the PES scans at large displacements (especially for ν_1).

Much, or even most, energy is contained in the higher-frequency modes, which are generally well described by the harmonic approximation. Unreasonable displacements along the low-frequency modes involving angular coordinates lead to significant errors, often compatible with the energy of high-frequency vibrations. As a result, the scaling factor deviates from 1.0 significantly but reflects neither the accuracy of sampling high-frequency modes, nor low-frequency modes. For the former, it is too far from 1.0, for the latter—too close to it. Consequently, by global scaling (Eq. (11)) it is impossible to fix unphysical displacements arising from large-amplitude vibrations. Moreover, this procedure will automatically spoil the quality of the low-amplitude vibrations sampling. Fortunately, this problem can be reduced, and in some cases solved completely, by performing the harmonic sampling in the curvilinear internal coordinates. A possible extension of the scheme can be designed based on the idea of mode-specific scaling, which goes beyond the scope of present work.

C. Trajectory modelling results

Finally, we have integrated two sets of 100 trajectories for acetone sampled at 300 K in the Cartesian and in the internal coordinates. The gradient-based velocity-Verlet integrator³⁰

TABLE I. Sampling of acetone at 300, 500, and 700 K using rectilinear and curvilinear normal coordinates at DFT/BLYP/6-311G** level of theory.

	Mean error (a.u.)		Maximal error (a.u.)		Mean deviation of k from 1.0		Average k	
	Rect.	Curv.	Rect.	Curv.	Rect.	Curv.	Rect.	Curv.
300 K	0.032	0.013	0.220	0.053	0.245	0.135	0.774	0.920
500 K	0.032	0.013	0.168	0.036	0.241	0.126	0.789	0.939
700 K	0.035	0.013	0.410	0.052	0.248	0.130	0.777	0.927

with the time step of 0.5 fs was applied, and the guess density was propagated using the time-reversible technique.³¹ The trajectories were run up to 5000 steps, or 2.5 ps.

It turned out that 45% of trajectories initialized using rectilinear modes resulted in the dissociation of one or several C–H bonds within 20 fs. On the contrary, only 5% of trajectories ended up in C–H bond breaking, when curvilinear normal modes were used for the sampling. This unphysical behaviour is the consequence of the coupling between the stretch and the torsion, especially strong in the case of Cartesian modes, as discussed above and plotted in Figs. 1 and 2. Typically, such problems are coped with by sampling several modes according to the classical velocity-distribution. However, this can be largely avoided by sampling in the internal coordinates.

V. CONCLUSIONS

We have shown that sampling low-frequency (large-amplitude) vibrations (especially those involving angular coordinates) using the conventional rectilinear normal coordinates is, even for small displacements, a very inaccurate procedure which cannot be improved by global coordinate scaling. The reason is that the scaling factor is defined to a large extent by high-frequency modes, which are well described by the (Cartesian) harmonic approximation, whereas the errors originate from low-frequency vibrations. On the other hand, using the curvilinear normal modes significantly improves the quality of the sampling in a simple black-box fashion, which requires almost no additional computational effort provided the HOPE internal-to-Cartesian back transformation is used.

An important application of the present approach would be sampling of rotational modes of individual molecules in molecular clusters. In this case, it may, however, be necessary to introduce new types of redundant internal coordinates, exactly as suggested in Ref. 13. Here, the new internal coordinates should correspond to the rotation of entire molecules. Because of this additional complication we leave molecular clusters as a topic for future work.

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